

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2456834.5962 +/- 0.0035 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.158 +/- 0.046
Transit depth (Rp/Rs)^2	2.51 +/- 1.47 [%]
Orbital Inclination (inc)	84.66 +/- 2.83
Airmass coefficient 1 (a1)	2.04 +/- 0.27
Airmass coefficient 2 (a2)	-0.23 +/- 0.4
Scatter in the residuals of the lightcurve fit is	11.91 %
Best Comparison Star	#1 - [625, 66]
Optimal Aperture	1.7
Optimal Annulus	3.0
Transit Duration (day)	0.0007 +/- 0.0018

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.158 $\pm$ 0.046 vs 0.1326 (deviation 0.55 σ)
Duration fit vs published	0.0007 d vs 0.0737 d (deviation 40.58 σ)
Mid-time O-C	-3105.6 min
Depth SNR	3.4
Residual scatter	11.91 %

Light curve

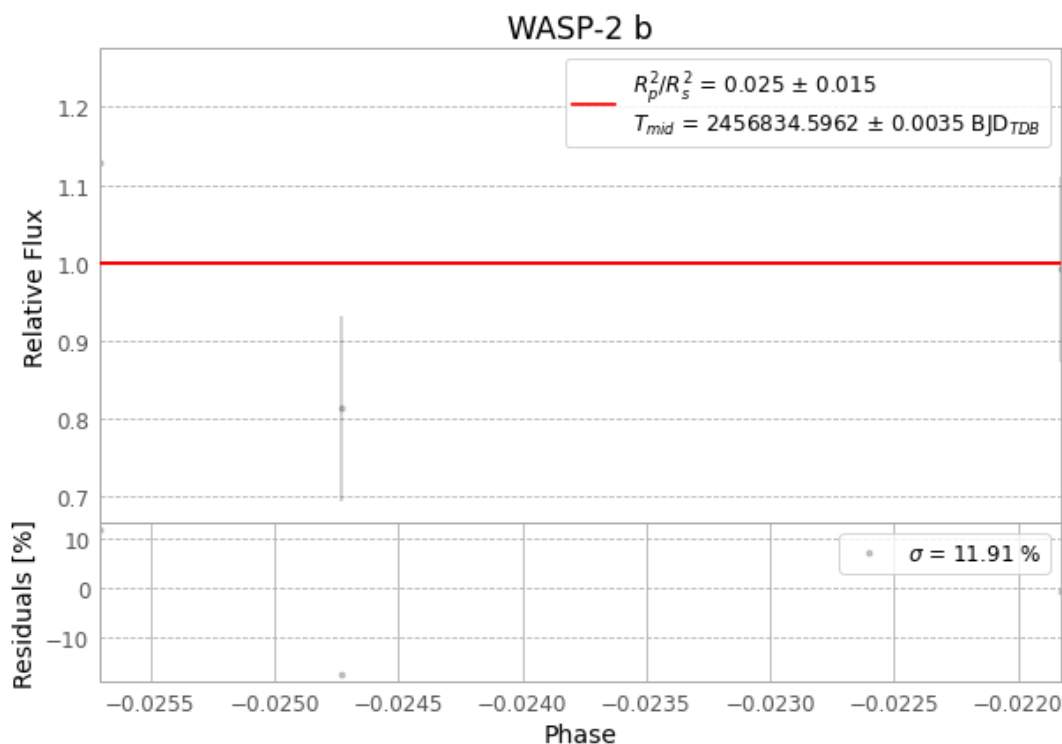


Figure 1 — FinalLightCurve_WASP-2 b_2014-06-27.png (SHA-256 d72df2cdcaf1ec1c...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [322, 227], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 bf7c8f256cbbfffb...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

70 FITS frames (46 MB), WASP-2140628043012.FITS → WASP-2140628075713.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-2-140628.log (14eb5e21a747...), FinalLightCurve_WASP-2 b_2014-06-27.png (d72df2cdcaf1...), FinalLightCurve_WASP-2 b_2014-06-27.csv (72efd9c2e946...)

Compute resources

Wall time 120.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-02 03:43Z.